



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 14 (Summary and Tutorial)

Topic: Discrete Random Variable, Binomial Distribution, Normal Distribution

Summary for Discrete Random Variable

Probability Distribution

A table or formula giving the values of $P(X=x)$ for every x in sample space is called the **probability distribution** of X .

For example, the experiment of tossing a fair coin 3 times where X is the number of heads obtained, the **probability distribution** of X is as follows:

x	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

The probability distribution of X satisfies the following:

1. $0 \leq P(X=x) \leq 1$ for all x in S .
2. $\sum_{x \in S} P(X=x) = 1$ where the summation is over all values of x in S .

Expectation of Discrete Random Variable

The expectation (or mean, or expected value) of a discrete random variable X taking values from a set S is given by $E(X) = \sum_{x \in S} xP(X=x)$.

In the example above, $E(X) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{3}{2}$

Independent Discrete Random Variables

Let X and Y be two discrete random variables taking on possible values $x_1, x_2, \dots$ and $y_1, y_2, \dots$ respectively. The random variables X and Y are said to be **independent** if for all i and j ,

$$P(X = x_i \text{ and } Y = y_j) = P(X = x_i)P(Y = y_j).$$

In the example above, if T is the event that the outcome of tossing a coin is head, then $P(X=0) = P(T_1=0, T_2=0, T_3=0) = P(T_1=0)P(T_2=0)P(T_3=0) = [P(T=0)]^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

This is possible given that T_1, T_2 and T_3 are independent.

Functions of a Discrete Random Variable

The expectation of $g(X)$, where g is a function of X , is denoted by $E(g(X)) = \sum_{x \in S} g(x) P(X = x)$.

In particular, $E(X^2) = \sum_{x \in S} x^2 P(X = x)$.

Variance and Standard Deviation of a Discrete Random Variable

The variance of a discrete random variable X is given by

$$\text{Var}(X) = E(X - \mu)^2 = E(X^2) - [E(X)]^2.$$

In the example above, $\text{Var}(X) = 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8} - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$

The standard deviation of X , denoted by σ , is defined as $\sqrt{\text{Var}(X)}$.

Properties of Expectation and Variance

(Note that these properties hold for discrete and continuous random variables)

Let X and Y be random variables and a and b be constants. We have

Expectation	Variance
(1) $E(a) = a$	(1) $\text{Var}(a) = 0$
(2) $E(aX) = aE(X)$	(2) $\text{Var}(aX) = a^2\text{Var}(X)$
(3) $E(aX \pm b) = aE(X) \pm b$	(3) $\text{Var}(aX \pm b) = a^2\text{Var}(X)$
(4) $E(aX \pm bY) = aE(X) \pm bE(Y)$	If X and Y are independent random variables, then (4) $\text{Var}(aX \pm bY) = a^2\text{Var}(X) + b^2\text{Var}(Y)$

Important Results

In general, $E(nX) = nE(X)$

$$E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n) = nE(X)$$

but $\text{Var}(nX) = n^2 \text{Var}(X)$

$$\text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n) = n \text{Var}(X).$$

Summary for Normal Distribution

Note that Normal random variable is a special continuous random variable. For continuous random variable, probability is calculated using the area under its probability density function.

In H2 Math syllabus, we just need to know how to use GC to evaluate these probabilities.

We also need to be aware that

$P(X < x) \neq P(X \leq x)$ for discrete random variable (Binomial included)

BUT

$P(X < x) = P(X \leq x)$ for continuous random variable

Normal Distribution

If a continuous random variable X follows a normal distribution, we write $X \sim N(\mu, \sigma^2)$,

where $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$.

Properties of a Normal Curve

Let $X \sim N(\mu, \sigma^2)$.

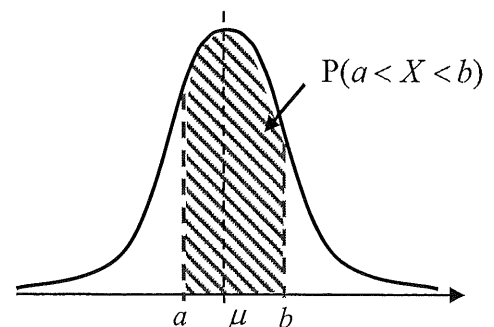
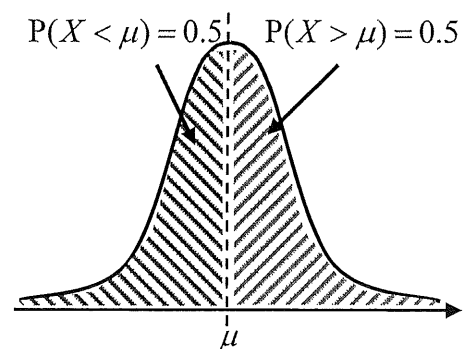
- (1) It is symmetrical about the line $x = \mu$.
- (2) The mean, median and mode are all equal to μ .
- (3) It approaches the x -axis as $x \rightarrow \pm\infty$.
- (4) Area under the graph gives the probabilities,

$$\text{i.e., } P(a < X < b) = \int_a^b f(x) dx$$

where $y = f(x)$ represents the probability density function of the normal curve.

Hence $P(a < X < b)$ is given by the area under the graph from $x = a$ to $x = b$.

- (5) Total area under the curve is 1.



$$(6) \quad P(\mu - \sigma < X < \mu + \sigma) \approx 0.68$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997$$

i.e., approximately 68%, 95% and 99.7% of the values drawn from a normal distribution lies within 1, 2 and 3 standard deviations of the mean respectively.

Standard Normal Distribution

Let $X \sim N(\mu, \sigma^2)$. The random variable Z , which is the standard normal variable, is defined by $Z = \frac{X - \mu}{\sigma}$. The standard normal distribution is $Z \sim N(0, 1)$.

The process of converting $X \sim N(\mu, \sigma^2)$ into $Z \sim N(0, 1)$ is known as **standardization**.

$$P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Remarks: Standardization is usually applied when there are unknown parameter(s) when we can't use the commands `normalcdf()` or `invNorm()` of our GC directly.

Using the Properties of Expectation and Variance of Random Variables, we have the following results for independent Normal Random Variables:

Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$ be **independent** random variables and a and b be constants. We have

$$(1) \quad X \pm Y \sim N(\mu_1 \pm \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$(2) \quad aX \pm b \sim N(a\mu_1 \pm b, a^2\sigma_1^2)$$

$$(3) \quad aX \pm bY \sim N(a\mu_1 \pm b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$$

Reminder (in revision 14 as well): Probability could easily be embedded in another Statistics question due to its nature. Read question carefully.

Revision Tutorial Questions**Source of Question: ACJC Prelim 9758/2017/02/Q6**

1. Alex and his friend stand randomly in a queue with 3 other people. The random variable X is the number of people standing between Alex and his friend.

- (i) Show that $P(X = 2) = 0.2$. [2]
- (ii) Tabulate the probability distribution of X . [2]
- (iii) Find $E(X)$ and $E(X - 1)^2$. Hence find $\text{Var}(X)$. [3]

Source of Question: JJC Prelim 9758/2017/02/Q5

2 The probability distribution of a discrete random variable, X , is shown below.

x	1	2
$P(X = x)$	a	b

Find $E(X)$ and $\text{Var}(X)$ in terms of a . [5]

Source of Question: MJC Prelim 9758/2017/02/Q6

3 The probability function of X is given by

$$P(X = x) = \begin{cases} (2x-1)\theta & \text{if } x = 1, 2, 3 \\ k & \text{if } x = 4 \\ 0 & \text{otherwise} \end{cases}$$

where $0 < \theta < \frac{1}{9}$.

- (i) Show that $k = 1 - 9\theta$. Find, in terms of θ , the probability distribution of X . [2]
- (ii) Find $E(X)$ in terms of θ and hence show that $\text{Var}(X) = 26\theta - 196\theta^2$. [3]
- (iii) The random variable Y is related to X by the formula $Y = a + bX$, where a and b are non-zero constants. Given that $\text{Var}(Y) = \frac{1}{3}b^2$, find the value of θ . [3]

Source of Question: AJC Prelim 9758/2017/02/Q10

- 4 Males and females visiting an amusement park have heights, in centimetres, which are normally distributed with means and standard deviations as shown in the following table:

	Mean (cm)	Standard deviation (cm)
Male	165	12
Female	155	σ

It is found that 38.29% of the females have heights between 150 cm and 160 cm.

- (i) Show that $\sigma = 10.0$ cm, correct to 3 significant figures. [2]
- (ii) Find the probability that the height of a randomly chosen female is within 20 cm of three-quarter the height of a randomly chosen male. State an assumption that is necessary for the calculation to be valid. [4]

The amount, \$ X , a visitor has to pay for a popular ride in the park is \$10 if the visitor's height is at least 120 cm but less than 150 cm, and \$ m if the visitor's height is 150 cm and above. If the visitor's height is less than 120 cm, he/she does not need to pay for the ride.

- (iii) Assuming that a visitor purchasing a ticket for the ride is equally likely to be a male or female, find in terms of m , the probability distribution of X . [3]

Given that the expected amount a visitor will pay for a ride is \$17.93, show that $m = 20.00$, correct to 2 decimal places. [1]

- (iv) Three visitors were randomly chosen. Find the probability that the total amount they paid for a ride together is more than \$40. [3]

Source of Question: HCI Prelim 9758/2017/02/Q5

- 5 The independent random variables X and Y are normally distributed with the same mean 7 but different variances $\text{Var}(X)$ and $\text{Var}(Y)$, respectively. It is given that $P(X < 10) = P(Y > 6)$.

- (i) Show that $\text{Var}(X) = 9\text{Var}(Y)$. [3]
- (ii) If $\text{Var}(Y) = 1$, find $P(X < 9)$. [2]

Source of Question: NJC Prelim 9758/2017/02/Q9

- 6 The number of days of gestation for a Dutch Belted cow is normally distributed, with a mean of μ days and a standard deviation of σ days. 8.08% of this cattle breed has a gestation period shorter than 278 days whereas 21.2% has a gestation period longer than 289 days. Find the values of μ and σ , giving your answers correct to 3 significant figures. [3]

- (i) Find the probability that the mean gestation period for thirty-two randomly chosen Dutch Belted cows is more than 287 days. State a necessary assumption for your calculation to be valid. [3]

For another cattle breed, the Jersey cow, the number of days of gestation is normally distributed with a mean of 278 days and a standard deviation of 2.5 days. During gestation, a randomly chosen pregnant Dutch Belted cow eats 29 kg of feed daily while a randomly chosen pregnant Jersey cow eats 26 kg of feed daily.

- (ii) Find the value of a such that during their respective gestation periods, there is a probability of 0.35 that the amount of feed consumed by a randomly chosen pregnant Jersey cow exceeds half of the amount consumed by a randomly chosen pregnant Dutch Belted cow by less than a kg. Express your answer to the nearest kg. [2]
- (iii) Calculate the probability that during their respective gestation periods, the difference between the amount of feed consumed by three randomly chosen pregnant Dutch Belted cows and four randomly chosen pregnant Jersey cows is more than 4000 kg. State clearly the parameters of the distribution used in the calculation. [3]

Source of Question: PJC Prelim 9758/2017/02/Q7

- 7 The masses, in kilograms, of black sea bass fish and red tilapia fish sold in a supermarket are normally distributed. The means and standard deviations of these distributions, and the selling prices, in \$ per kilogram, are given in the following table.

	Mean Mass (kg)	Standard Deviation (kg)	Selling Price (\$ per kg)
Black sea bass fish	1.10	0.20	12
Red tilapia fish	0.55	0.05	9

- (i) Ayden bought 2 black sea bass fish and 3 red tilapia fish. Find the probability that he pays more than \$40. State an assumption needed in your calculation. [4]
- (ii) Five red tilapia fish are randomly chosen. Find the probability that the fifth red tilapia fish is the third red tilapia fish weighing less than half a kilogram. [3]

Source of Question: DHS Prelim 9758/2018/02/Q6

- 8 A test consists of 15 multiple choice questions, where each question has n possible options, of which only one is correct. A student took the test by randomly choosing an answer to each question. It is known that the probability of answering exactly 3 questions correctly is the same as the probability of answering exactly 4 questions correctly.

(i) By forming an equation in terms of n , find the value of n . [3]

Each correct answer is awarded 3 marks and each incorrect answer carries a penalty of 1 mark. The score is the total marks awarded based on the number of correct and incorrect answers.

(ii) Find the expected score, s , obtained by the student. [3]

(iii) Find the probability that the score obtained by the student is within 4 marks of s . [2]

Source of Question: EJC Prelim 9758/2018/02/Q7

- 9 When Mr. Lee sends a text message to any of his students over the weekend, he gets a reply, on average, about 6 out of 10 times. On a particular weekend, Mr. Lee sends a text message to 25 students.

(i) State, in the context of this question, two assumptions needed to model the number of students that reply by a binomial distribution. [2]

(ii) Explain why one of the assumptions stated in part (i) may not hold in this context. [1]

Assume now that these assumptions do in fact hold. Stating the distribution clearly,

(iii) find the probability that at least half of the students reply. [3]

It is given instead that the probability of a randomly chosen student replying is p . Find the least value of p such that there is at least a 90% chance of more than 20 students replying to the text message. [3]

Source of Question: IJC Prelim 9758/2018/02/Q5

- 10 The random variable X has the distribution $B(25, p)$, where $0 < p < 1$. Given that $P(X \leq 1) = 0.15$, write down an equation for the value of p and find this value numerically. Hence find $\text{Var}(X)$. [4]

Source of Question: NYJC Prelim 9758/2018/02/Q6

- 11** A game is played using a fair six-sided die, a pawn and a simple board as shown below.

S	1	2	3	4	5	E
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Initially, the pawn is placed on square S. The game is played by throwing the die and moving the pawn in the following manner:

S 1 2 3 4 5 E 5 4 3 2 1 2 3 4 5 E.....

Thus, for example, if the first and second throw of the die gives a “5” and “4” respectively, the final position of the pawn will be on square “3”.

The game will stop when the pawn stops at square E.

Let X be the random variable denoting the number of throws of the die required to move the pawn such that it stops at square E.

- (i) Show $P(X = 2) = \frac{5}{36}$. [1]
- (ii) Find the probability that more than two throws of the die are needed for the pawn to stop at square E given that the first throw of the die gives an even number. [3]

It is now given that for each game, a player has a maximum of 3 throws of the die and a special prize is given to any player who uses not more than two throws for the pawn to stop at square E.

- (iii) Find the probability of a player winning a special prize in at least three but not more than eight games out of ten games. [3]
- (iv) Find the least number of games needed so that the probability of winning at least a special prize is at least 0.998. [3]

12 A graphing calculator is not to be used for this question.

Joel the student lives at Holland Village and he is considered late if he reaches the school parade square after 0740. He catches the 0630 bus to school with probability p , or the 0650 train with probability $1 - p$. Measured in minutes, the journey time for the bus follows a normal distribution with mean 40 and standard deviation 12. The journey time for the train follows another normal distribution with mean 30 and standard deviation 4.

Journey times from the bus-stop where he alights to the parade square in school follows a normal distribution with mean 18 and standard deviation 5. The journey time from the train station where he alights to the parade square in school also follows a normal distribution with mean 10 and standard deviation 3.

Show that Joel is more likely to be late for school if he takes the bus, writing down any assumptions necessary in your calculations. [6]

Let A be the probability Joel is late if he takes the bus, and B the probability he is late if he takes the train. Over a period of M days, where M is large, Joel noted that in 60% of the occasions when he was late, he had taken the bus. Obtain an estimate of p in terms of A and B . [3]

- 13** 3 bar magnets are placed randomly end-to-end in a straight line. If adjacent magnets have ends of opposite polarities facing each other, they join together to form a single unit. If they have ends of the same polarity facing each other, they stand apart. Let L be the number of separate units.

(i) Show that $P(L = 1) = P(L = 3) = \frac{1}{4}$. [2]

(ii) Hence find the exact values of $E(L)$ and $\text{Var}(L)$. [4]

Instead of just 3 bar magnets, there are now 10 bar magnets placed randomly end-to-end in a straight line. We say there is a change in polarity when ends of opposite polarities face each other. Let S be the total number of changes in polarities.

(iii) State a necessary assumption for S to follow a binomial distribution. Assuming the condition holds, state the parameters of S . [3]

(iv) Hence find the expected number and variance of the number of separate units. [3]

